Sanchit Kukreja

+91-8178-898-360 | ksanchit778899@gmail.com | LinkedIn | GitHub | Portfolio

Education

Bennett University — Uttar Pradesh, India
Bachelor of Technology (B.Tech), Computer Science in AI/ML; CGPA: 8.35 — *2024 – 2028*

VSPK International School — New Delhi, India
CBSE Class X: 88% (2021); Class XII: 88% (2023) — *2023*

Technical Skills

Languages: Java, Python, C++, JavaScript
Backend/Frameworks: FastAPI, Spring Boot, Spring Security, Flask, Streamlit
ML & Data Science: Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn
Computer Vision: YOLO, OpenCV, Image Classification, Object Detection
Generative AI: LLMs, Prompt Engineering, Embeddings, RAG, Vector Databases, LangChain
Databases: MySQL, MongoDB
Developer Tools: Git, GitHub, Docker, Kubernetes, Postman, VS Code, Jupyter
Core CS: Data Structures & Algorithms, OOP, Operating Systems, DBMS, Computer Networks

Projects

FixMyCampus – University Issue Tracking Platform | *Node.js, SQLite*

- Developed a campus issue reporting and resolution platform using Node.js and SQLite, enabling students and faculty to report maintenance issues and track resolution progress.
- Implemented role-based access control for students, faculty, administrators, and departments, with permissions for ticket creation, assignment, progress updates, and resolution.
- Built ticket lifecycle features including department assignment, progress timelines, priority filtering, escalations, resolution feedback, and reopen requests; implemented cookie-based authentication, email verification, password reset, and automated end-to-end smoke tests.

Intrusion Detection System (IDS) Dashboard | *Flask, Scikit-learn, Random Forest, NSL-KDD*

- Developed a Flask-based intrusion detection dashboard using a Random Forest classifier trained on 148,517 NSL-KDD records with 41 input features across six attack classes.
- Built a machine learning pipeline with categorical encoding, feature scaling, intrusion classification, attack probability estimation, and risk categorization.
- Integrated system telemetry using `psutil`, with network activity analytics, connection monitoring, detection alerts, and historical visualization.
- Implemented CSV-based batch prediction and attack simulation capabilities to analyze network activity and evaluate detection behavior.

AI-Driven Runoff Prediction for Flood Forecasting and Climate Resilience | *Python, Scikit-learn, Random Forest*

- Built a 3-day-ahead runoff forecasting pipeline using 11,323 daily hydrometeorological observations from the Kasol Basin (1979–2009), engineering 22 hydrological, lagged-rainfall, seasonal, and meteorological features.
- Trained and evaluated a log-transformed Random Forest using chronological holdout validation, achieving 124.32 CUMEC RMSE and 0.867 R^2 across 1,456 test observations from 2006–2009.
- Evaluated extreme-flow predictions, recording 272.51 CUMEC RMSE on the top 10% of test flows and +2.95% PBIAS on the overall test set.
- Implemented an end-to-end prediction pipeline with discharge normalization, feature preprocessing, and inverse log transformation.

Achievements

- Recognized as Computer Science Topper in Class XII (2023–24) at VSPK International School.
- Ranked among the Top 100 teams in the SIH 2026 University Hackathon.
- Ranked among the Top 10 teams in Hackacinno.
- Solved 800+ coding problems across LeetCode, CodeChef, and GeeksforGeeks.